

Three ideas. One capstone.



uofrEats

Campus ordering and pickup



AI Learning Environment

Learning with your course files



AI-Assisted Drones

Crop scouting and fire detection

What is uofrEats?

DEFINITION + WHY

A campus food-ordering platform

Students order from campus vendors, pay per order and choose an available pickup time.

WHY THIS PROBLEM?

Students have short breaks. Long ordering lines can cost them a meal and cost vendors a sale.



Campus pickup designed around class breaks. No meal plan required.

Who is uofrEats for?

WHO

First people interviewed: restaurant owners/managers and students

“During peak rush hours, some students leave without buying food because the line is too long.”

Manmeet, Da India Curry House Xpress manager

Interview summary, paraphrased from team notes

STUDENTS

Commuters and residents who need food between classes

RESTAURANT TEAMS

Owners, managers and staff handling the lunch rush

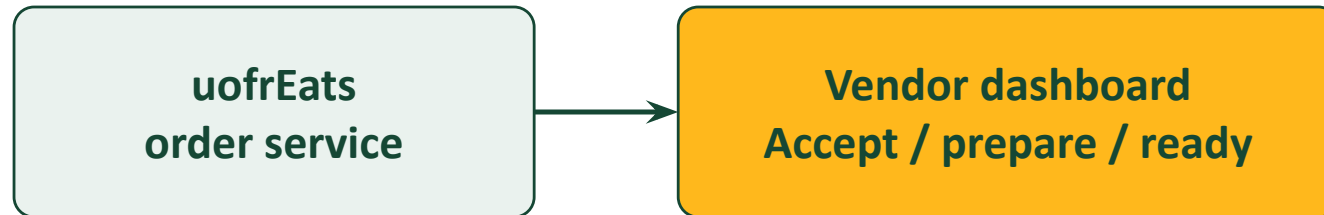
Two interfaces, two experiences

WHAT + HOW

1. Student interface: order and collect

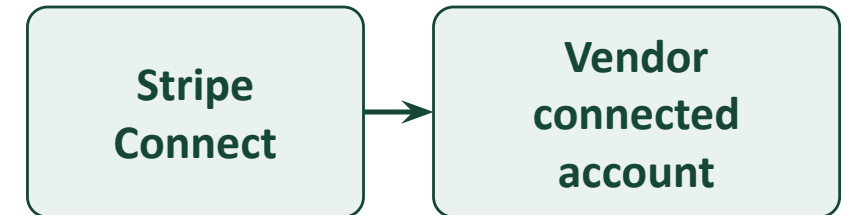


2. Vendor interface: manage preparation



Orders and status updates

Card payment routing



Restaurant receives funds

Pickup capacity follows the kitchen. Ordering ahead must not overload it.

uofrEats value and competitors

VALUE + STRATEGY

Selected comparison	Existing experience	uofrEats proposed focus
UofR Dining / UR OneCard	Dining information and prepaid meal-plan payments	Order and pay per purchase, including students without meal plans
Uber Eats	Many local restaurants, delivery and scheduled pickup	A campus-only vendor view shaped around class breaks and kitchen capacity

PROPOSED UNIQUE VALUE

One campus ordering space, with pickup capacity controlled by each vendor

Strategy: start with one vendor, test pickup reliability, then expand campus coverage.

Coverage check: the campus curry outlet is on Uber Eats. Other outlets need verification.

uofrEats research phases

HOW / PDSA

01

PLAN

Interview students and map Manmeet's peak-hour workflow.

02

DO

Test an ordering prototype and a vendor order screen.

03

STUDY

Compare pickup wait, readiness and staff handling time.

04

ACT

Adjust slot capacity, order acceptance and pickup details.

QUESTION TO TEST

Can scheduled orders reduce waiting without increasing staff workload?

First prototype

One vendor, short menu, pickup slots, test payments and order status.

AI Learning Environment

DEFINITION + WHY

A student-designed learning workspace

Learn from your local course files through explanations, conversation, practice and planned review.

WHY THIS PROBLEM?

Repeated uploads and switching between tools can interrupt a study session.



Goal: a smoother path from reading a PDF to understanding the concept

Who is the learning workspace for?

WHO

Student interview: Linda

“Asking ChatGPT questions can feel awkward when I have to upload files for context. It would be smoother if it could see the file I am studying and explain it.”

Linda, student interview participant

Interview summary, paraphrased from team notes

PRIMARY USERS

Students learning from PDFs on desktop and iPad

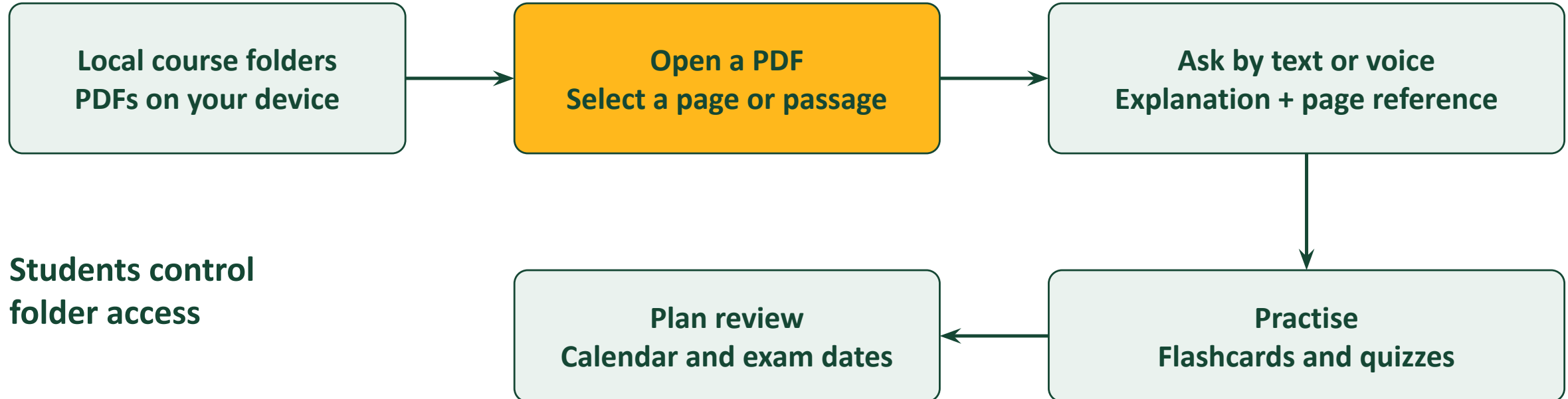
DESIGN PARTNERS

Students testing explanations, practice and ease of use

A study session with your material

WHAT + HOW

Proposed workflow



Students control
folder access

Original files stay on the device.

Research question: which on-device AI can deliver useful explanations on desktop and iPad?

Learning value and competitors

VALUE + STRATEGY

Existing option	What it already offers	Our research focus
NotebookLM	Source-based explanations, quizzes and flashcards	Less setup around local course folders
Goodnotes AI	AI questions across notes/PDFs and quizzes	A complete study-to-review workflow
Adobe Acrobat AI	PDF questions, citations and mobile voice	Practice and course planning together

PROPOSED UNIQUE VALUE

Help with the file in front of you, followed by practice and planned review

Adoption question: would students pay for a smoother workflow and local file control?

Learning environment research phases

HOW / PDSA

01

PLAN

Interview students and observe one real PDF study session.

02

DO

Prototype page explanation, source references and a short quiz.

03

STUDY

Test setup time, explanation accuracy and delayed recall.

04

ACT

Improve the study flow and narrow device requirements.

QUESTION TO TEST

Does this workflow improve understanding and reduce context-setting effort?

First prototype

One desktop workflow and one course. Extend to voice, calendar and iPad.

AI-Assisted Drones

DEFINITION + WHY

Crop disease screening and fire detection in Saskatchewan



Existing hardware shown as a concept reference

DEFINITION

An aerial system that flags crop problems and signs of fire for human review.

WHY

Large areas take time to inspect.
Useful alerts need a location to investigate.

Who could use the drone system?

WHO



Farmers and agronomists

Need to know where to inspect crops and which suspicious areas to prioritize.



Land and fire-monitoring teams

Need timely observations with a location and imagery they can review.

Stakeholder interviews are next. Crop scouting leads the first research phase.

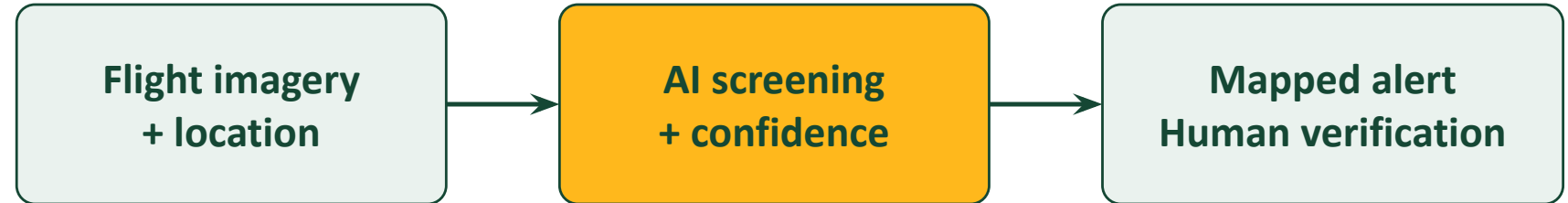
Aerial observations and human review

WHAT + HOW



FPV form factor reference

Existing DJI Avata hardware



CROP DISEASE SCREENING

Flag suspicious crop areas
for ground inspection

FIRE DETECTION

Flag smoke or thermal cues
for a person to review

Manual control stays available

Pilot-assisted flight first. Autonomous missions are later scope.

Drone value and competitors

VALUE + STRATEGY

Existing option	What it already offers	Our research focus
PIX4Dfields / DroneDeploy	Field mapping and plant-health analysis	A narrow condition with validated alerts
DJI Mavic 3M	RGB and multispectral crop imaging	Software integration and human review
DJI Matrice 4T	Thermal imaging and inspection tools	A small fire/smoke screening experiment

PROPOSED UNIQUE VALUE

Locations to investigate, with the image and confidence behind each alert

Business question: can a focused scouting service justify its equipment and review costs?

Drone research phases

HOW / PDSA

01

PLAN

Interview an agronomist and define one crop condition and sensor.

02

DO

Test recorded imagery, one detector and a mapped alert.

03

STUDY

Measure false alerts, missed cases, location error and latency.

04

ACT

Improve field reliability, then assess a supervised flight trial.

QUESTION TO TEST

Can a narrow detector produce alerts a real user can trust and act on?

First prototype

Crop scouting first. Fire detection starts as a separate imagery experiment.

Comparing the three alternatives

SELECTION

Decision factor	uofrEats	AI Learning	AI Drones
Early evidence	Manmeet interview	Linda interview	Interviews pending
First prototype	One-vendor pickup workflow	One-PDF study session	One detector and mapped alert
Main dependency	Vendor access and kitchen workflow	Device performance and student testing	Labelled imagery, hardware and field access
Biggest unknown	Pickup reliability and adoption	Learning benefit and ease of use	Reliability on unseen field conditions

Selection criteria: validated need, access to users and a scope two people can deliver

Any questions?

Aubin Chriss Izere / 200490675

Chop Kur / 200497265

<https://github.com/Binborg1/Final-2027.git>